PRERANA ACHAR

M.Sc. Machine Learning Student at the University of Tübingen
Master's Student Intern at the Max Planck Institute for Intelligent Systems

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PUBLICATIONS

2025 **BEDLAM2.0: Synthetic Humans and Cameras in Motion** **NeurIPS 2025 (Oral)**
J. Tesch, G. Becherini, **P. Achar**, A. Yiannakidis, M. Kocabas, P. Patel, M. J. Black. *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2025. **Selected for oral presentation.**
Project page • arXiv:2511.14394

RESEARCH EXPERIENCE

09/24 - Current **Max Planck Institute for Intelligent Systems - Tübingen, Germany** **Master's Student Intern**
- Student intern at the Institute of Perceiving Systems supervised by Dr. Michael J. Black
- Worked on the footwear component of the BEDLAM dataset.
- Current work includes: Generating heel-adapted body motion and physics driven cloth simulation.

03/23 - 05/24 **Delft University of Technology - Delft, Netherlands** **Bachelor's Thesis**
- Thesis on Material-Aware Multilayered Garment Draping for Digital Humans, supervised by Prof. Chirag Raman.
- Built a graph-neural-network model that drapes garments over a 3D human mesh supervised in a data-free way using physics potentials.

09/23 - 01/24 **Universidad Rey Juan Carlos (URJC) - Madrid, Spain** **Research Project**
- ShoeMPL, supervised by Prof. Dan Casas: a model that generates textured 3D shoe and body textures for the parametric body model SMPL from text inputs.

2022 **Text to Textile: Generation of Optimized 3D Garments from Textual Prompts** **Independent Research**
- Text-to-3D clothing generator for digital humans that uses text prompts and template meshes to guide mesh deformations and generate clothing for fantasy avatars.
- Includes a Blender add-on to tweak garment parameters (length, texture, sleeve height, skirt, jacket, etc.) from user-entered prompts.

EDUCATION

10/25 - Current **University of Tübingen - Tübingen, Germany** **M.Sc. Machine Learning**
Courses: Probabilistic ML, Statistical ML, Deep Learning, 3D Vision, Advanced Computer Graphics, Development of the Brain and Mind, Introduction to General Relativity

10/20 - 06/24 **Dwarkadas J. Sanghvi College of Engineering - Mumbai, India** **B.Tech in IT (Honours)**
GPA - **9.42/10**
Bachelor's thesis conducted with the Pattern Recognition & Bioinformatics group at TU Delft
Relevant courses: Machine Learning, Advanced Machine Learning, Image Analysis and Computer Vision, Probability and Statistics, Statistical Analysis

INDUSTRY EXPERIENCE

06/23 - 08/23 **JP Morgan Chase & Co.** **Software Engineering Intern**
- Built abstract-syntax-tree parsing and translation of an internal database language to Azure Databricks SQL.

07/22 - 08/22 **Net Tech Services India Pvt Ltd** **Machine Learning Intern**
- Improved a CNN-based OCR model from 94% to 98% accuracy for extracting handwritten English and Hindi data from cheques.

HOBBIES

I also enjoy playing different tabletop RPGs, art, developing and playing video games, table tennis and the violin.